

Kwangmin Ki

kwangminki@korea.ac.kr | Homepage | Google Scholar | LinkedIn

Research Interests

LLM Reasoning, Reliability, Efficiency

Education

2021 – **B.S., Computer Science & Life Sciences, Korea University**

Major GPA: 4.30/4.5 | Total GPA: 4.15/4.5

2015 – 2018 International School of Prague

Publications

[P1] **Efficient Reward Hacking Detection using Answer Likelihood under Truncated Reasoning**

Kwangmin Ki, Minju Gwak, Jaehyung Kim

Under Review (ICLR 2027)

[C3] **CPR for LLMs: Critical-Point Routing against Catastrophic Forgetting in Domain Adaptation**

Kwangmin Ki, Yunhun Nam, Jongheon Jeong, Jaehyung Kim

EMNLP 2026 (Main, 🏆Oral)

[C2] **How Instruction Hierarchy Breaks Under Pairwise Multi-Conflict Pressure**

Yerin Hwang, Jiwon Moon, Changho Lee, **Kwangmin Ki**, Taegwan Kang, Jinsik Lee, Honglak Lee, Kyomin Jung

EMNLP 2026 (Main)

[C1] **When to Ensemble: Identifying Token-Level Points for Stable and Fast LLM Ensembling**

Heecheol Yun, **Kwangmin Ki**, Jung Hyun Lee, Eunho Yang

ICLR 2026 (Poster)

Research Experience

2026 – **Undergraduate Researcher**, ML3 Lab, Yonsei University

Advisor: Jaehyung Kim

- Proposed critical-point routing to mitigate forgetting in LLM domain adaptation [C3]
- Proposed efficient reward hacking detection via likelihood under truncated reasoning [P1]

2025 **Research Intern**, Language Lab, LG AI Research

Mentors: Youbin Ahn, Yerin Hwang

- Contributed to MIH-Bench, a multi-conflict instruction hierarchy benchmark for LLM [C2]
- Built a large-scale Excel SFT dataset for EXAONE 4.0-32B, approaching Qwen3-235B

2025 **Undergraduate Researcher**, MLI Lab, KAIST

Advisor: Eunho Yang

- Contributed to a selective token-level ensembling method for stable and efficient LLM [C1]

Extracurricular Activities

2024 – 2025 **President**, KUBIG (Korea University Data Science Society)

Advisor: Sungbin Lim

- Instructed the 2025 Summer NLP Bootcamp for undergraduate and graduate students
- Co-organized hackathon with partner AI societies

2024 **TechCamp Korea Workshop on Responsible AI**, U.S. Embassy

- Participated in a Responsible AI workshop led by Google and Intel engineers

2024 **K-Data Science Hackathon**, National Research Foundation of Korea

- Received the Excellence Award for a multi-module AI lesion diagnosis system

2023 – 2024 **Mandatory Military Service**

Personal Projects

- 2025 **Verifier-Guided Soft Contrastive Fine-Tuning for Retrieval-Augmented Generation**
- Proposed unsupervised retriever fine-tuning with zero-shot verifier-based re-ranking
 - Selected as an outstanding paper in the NLP course
- 2025 **Domain-Optimized RAG via Embedding Customization and Structured Retrieval**
- Built a title → chunk → sentence retrieval pipeline with domain-tuned embeddings
- 2024 **Kaggle Machine Learning Projects**
- Reached Kaggle Expert (top 5%)

Technical Skills

Languages	Python, C, Bash
Frameworks	PyTorch, Hugging Face (Transformers, PEFT, TRL), verl
Training	SFT, LoRA, RL (PPO, RLOO, GRPO), Distributed Training (FSDP)
Inference	vLLM, Speculative Decoding, KV Caching, Quantization
Evaluation	LLM-as-a-Judge, Benchmark Construction

Language Proficiency

Korean	Native
English	TOEFL 110/120
German	Goethe-Zertifikat B1